

The business analytics skills employers are looking for

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Abstract

Business analytics is a fast-growing field that requires a combination of technical, analytical, and communication skills. This article aims to identify the most sought after skills for business analytics jobs based on a content analysis of over 2600 online job postings. The results show that the top skills include analytics, communication, visualization, data science, business requirements, and problem-solving. The article also discusses the implications of these findings for curriculum design and career development. Aside from the standard battery of statistics classes, we argue that every analytics curriculum should contain components teaching statistical communication, visualization, databases, and programming as well as include a capstone project.

KEYWORDS

academic areas, analytics, content areas, curriculum design, pedagogy, undergraduate education

1 | INTRODUCTION

The demand for business analytics skills has increased significantly in recent years as more organizations recognize the value of data-driven decision-making. According to the US Bureau of Labor Statistics, the fields closely related to business analytics are expected to grow much faster than average: employment of operations research analysts is projected to grow 23% from 2023 to 2033,¹ and employment of data scientists is expected to grow 36% in that same time frame.²

It is important to identify which specific skills employers are looking for in business analysts. To succeed in the field of business analytics, professionals need to possess a combination of technical and nontechnical skills and competencies.³ Technical skills refer to the ability to use various methods and tools to collect, process, analyze, and present data, such as programming languages, statistical software, data visualization tools, and databases. Nontechnical skills refer to the ability to communicate, collaborate, problem-solve, and think critically and creatively in the context of business analytics.

Several studies have attempted to identify the most demanded skills for business analytics jobs based on online

job postings, surveys, or interviews. Chen et al. (2012) discussed the evolution of business intelligence and analytics, the role of big data, and their impact on organizations. The authors emphasized the importance of big data analytics in creating value for organizations and highlighted the need for organizations to develop analytics capabilities to create value from data. Overall, the literature underscores the importance of business analytics in improving organizational performance and highlights the need for further research to better understand the factors that influence its adoption.

Verma et al. (2019) conducted a content analysis of the skills listed in online job advertisements for business and data analysts, business intelligence analysts, and data scientists. They present ranked lists and comparisons of the relevant skills belonging to different positions. The authors observed that decision-making, organization, communication, and structured data management are key to all job categories. Their analysis shows that technical skills like statistics and programming skills are in most demand for DAs.

Our present study significantly enhances the literature on curriculum development within business analytics programs by pinpointing the skills currently in high demand across various job categories. Higher education institutions can use the study's outcomes to tailor their curricula to the skills employers most value, ensuring that their graduates are equipped with the most relevant and sought-after skills in the cutting-edge field of business analytics. The results of our analysis also hold practical significance for job seekers aiming for employment as business analysts. The clarity provided by our

¹ US Bureau of Labor Statistics' Occupational Outlook Handbook: <https://www.bls.gov/ooh/math/operations-research-analysts.htm>. Business analytics is not listed in the BLS's handbook, so we report results for these two analytics-adjacent careers.

² US Bureau of Labor Statistics' Occupational Outlook Handbook: <https://www.bls.gov/ooh/math/data-scientists.htm>

³ McKinsey Global Institute. 2011. "Big Data: The Next Frontier for Innovation, Competition, and Productivity." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/big-data-the-next-frontier-for-innovation>



delineation of skillsets enables job seekers to build a more targeted portfolio of skills to include in their applications.

To navigate through the comprehensive insights and contributions of this study, our article is organized as follows. We begin by discussing the related literature. We next describe the process by which our data are collected and analyzed. We then identify the most valued skills among business analytics job postings on LinkedIn.com. After this, we sharpen our focus to explore more specific questions such as the type of visualization skills that are demanded, programming skills, and the compatibility of other majors with analytics. We conclude by identifying various skills clusters and close with a discussion on the curricular implications of these findings.

2 | RELATED LITERATURE

We survey two strands of literature. The first analyzes the current analytics and data-centered curricula. This literature examines what is currently being taught. The second strand examines job descriptions for the data-centered occupations, to see what should be taught.

Gorman and Klimberg (2014) reviewed 34 master's and 10 undergraduate business analytics programs. The master's programs at the time were dominated by the statistics curriculum. This was followed by operations research (26%), databases (approximately 21%), and finally, business skills. They found some inertia in the curricula of master's and undergraduate analytics programs. Wherever business analytics programs were modifications of existing information systems (IS) or operations research (OR) programs, their curricula were more heavily skewed toward IS or OR topics. Programs that were started *de novo* emphasize these less. Gorman and Klimberg also found that traditional statistical software such as SAS dominated R. Our research below reveals this is no longer true.

Wixom et al. (2014) surveyed a group of professors, students, and industry professionals at the Business Intelligence Congress. They reported that industry especially values foundational skills such as, in decreasing order, (1) communication, (2) SQL, (3) basic statistics, (4) data management, and (5) business knowledge. Almost all academic programs offer classes in data mining/predictive analytics, introductory BA/BI, statistics, quantitative analysis, and modeling. Data management and data warehousing are also valued.

Aasheim et al. (2015) tried to identify differentiators between the data analytics (DA) and data science (DS) curricula. A small-scale study, they looked at 8 DA and 5 DS undergraduate programs and their course descriptions. The analytics programs were mostly taught in business schools, and DS in computer science programs. The former emphasizes databases and data warehousing more than do the DS programs, which focus more on algorithms and mathematics. Perhaps showing their roots as outgrowths of decision science programs in business schools, the analytics programs emphasize decision-making far more than the DS programs. Essentially, business analysts use data to identify and solve

business problems; data scientists use data to create new data-based products.

Ceccucci et al. (2020) audited a random sample of 225 AACSB-accredited programs, to investigate the overlap between business analytics and information systems degree programs. They find a large degree of overlap between the two majors. A common set of foundational courses—Intro to Business Analytics, Predictive Analytics, and Database Management—are prevalent among both programs. Decision modeling and an Intro to Programming round out the top five required classes. Surprisingly, from our perspective, Visualization ranks eighth. We find that visualization has become a defining feature of business analytics. Therefore, Ceccucci et al. (2020) reveal a gap in the curricula of many AACSB-accredited schools.

The second strand of literature examines job descriptions for analytics jobs. This research seeks to understand which skills are being demanded, how these demands have changed over time, and which skills differentiate one type of job, or job candidate, from another.

Radovilsky et al. (2018) examined whether there are any skills that differentiate data analytics and data science jobs. They examined 1050 job posts from various online job sites. Thus, it is a strong complement to Aasheim et al. (2015), which did the same for course offerings. They find that analytics jobs emphasize SQL far more than DS jobs. They also prioritize personal productivity and office software (Excel and PowerPoint). Proficiency with Tableau was a defining skill for analysts rather than data scientists. In fact, Tableau and visualization were not listed among the top 20 skills for data scientists. On the other hand, the data science jobs emphasized more mathematical proficiency and software development skills.

Some job postings emphasize that they are particularly recruiting business majors; others are recruiting from non-business technical majors like computer science, mathematics and engineering; other job postings are major-agnostic. Deng et al. (2016) investigate whether businesses list different skills when recruiting from these two applicant pools, ultimately finding only small differences. When recruiting business majors, they want them to have algorithms and mathematics. When recruiting STEM majors, they want them to have business knowledge. The implication is that firms want well-rounded candidates for business analytics jobs. These jobs include a mix of proficiencies including business domain knowledge, algorithms, and programming.

Dong and Triche (2020) analyze 9000 Indeed.com job postings for entry-level “business intelligence” and “data analytics” positions. Their findings highlight the critical need for a well-balanced curriculum that incorporates not only technical skills like Python, Excel, and Tableau but also soft skills like communication and collaboration. Their dataset spans several years, 2014–2018, so they are able to identify some trends. First, they document a dramatic rise in demand for Python Tableau, R and visualization. Other skills, such as MS Access, SAP, and Cognos, have seen decreasing demand. Database, visualization, and programming skills were highly



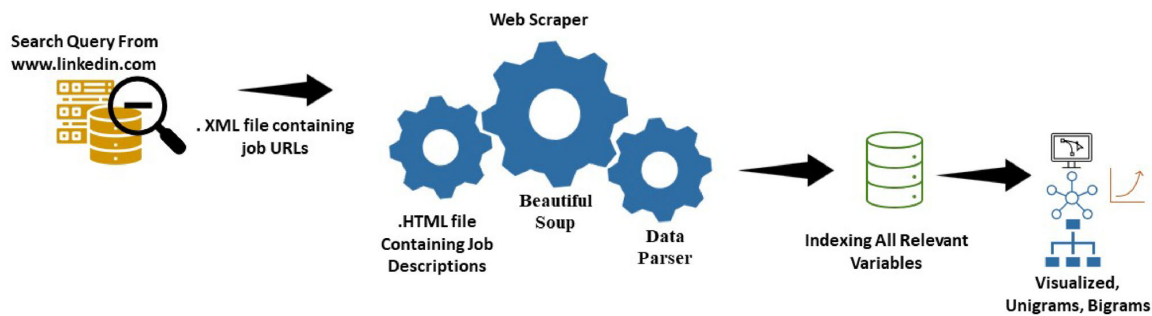


FIGURE 1 Summary of the methodology.

demand, and even more so at the end of their sample. Proficiency in databases is demanded more than proficiency in business intelligence platforms (such as Tableau and Power BI).

More recently, Umamaheswaran et al. (2023) examined online Business Analytics job postings in India from Indeed.com. Their analysis reveals several clusters of skills demanded of Indian business analysts. They identify a large amount of growth in demand for Python, SQL, Tableau, and data science. General visualization skills are less important in their sample than platform-specific proficiency in Tableau. Teamwork is the dominant “soft skill.” While their paper shares some similarities to ours, their results may be specific to the Indian market and may not generalize to the United States. Our methodology, while less technologically sophisticated, is also more transparent.

3 | METHOD

To collect data for this study, we searched for business analytics jobs on LinkedIn.com, a popular online job board. We used query phrases such as “business analytics,” and job titles containing these phrases were searched for three months between September 2023 and December 2023. We downloaded the job title, job descriptions, job types, and skills lists using Python 3.7.3, then created a parser to extract complete job descriptions from the 2676 job URLs collected from the search results, using HTML tags. The full job description contains relevant information such as position type, salary, full-time/part-time, company name and location, number of employees, and skills, among other details. We then indexed all relevant job information and downloaded it as a CSV file. This data will be used in the next section to analyze the relative importance of skills and skill categories for different job categories (Figure 1 provides a graphical “Summary of the Methodology”).

We used Stata 18.0 to code the job postings based on the skills listed in each LinkedIn posting’s skills section, perform content analysis of the job postings, and generate the frequency tables used throughout the article. We used VOSviewer version 1.6.0 for cluster analysis of the skills in each job post.

Several challenges were encountered during data collection and analysis. Firstly, some companies post one posting for multiple hires. This could lead to an underrepresentation of the demand for specific skills associated with those positions. On the other hand, sometimes the same job is posted several times until the position is filled. This would overrepresent the job and its associated skills. Where possible, duplicate job postings were systematically removed based on key identifiers such as the job title, company name, and date posted. Ultimately, it is impossible to completely correct these possibilities; we merely hope that the two tendencies offset each other. We treat each job posting as a single unit of analysis, irrespective of the number of openings it might represent.

Another challenge was that some skills, though identical or similar, were listed differently in either the skills section or the job description, contributing to ambiguity during the coding process. To address this, we often report results “as is,” and then “as though” similar skills are grouped together “by hand,” as it were. That is, we categorize similar or synonymous skills into unified categories. This categorization necessarily relies on our subjective judgment.

4 | RESULTS

4.1 | Most demanded skills

Which skills are most in demand for business analytics jobs? A simple approach would be to list the skills exactly as given in the job posting’s skills section. Unfortunately, the task is not so simple since a. In order to account for synonyms, we are forced to combine similar skills, and this requires a bit of judgment and subjectivity.

For example, we combine “analytics,” “business analytics,” “business analysis,” “data analytics,” and “data analysis” into one category. We treat visualization as a single skill, combining under one “visualization” heading all skills that include the following keywords: “dashboard,” “visualization,” “Power BI,” and “Tableau.” We combine “presentations” and “presentation skills,” and also “communication,” “communication skills,” and “oral communication.” Finally, we combine “databases,” “relational databases,” and all other skills that include the “database” string.



TABLE 1 Frequency of skills for business analytics jobs.

Rank	Skills	Frequency
1	Analytics (bus, data)	1651
2	Communication skills	828
3	Visualization	606
4	Analytical skills	588
5	Business requirements	424
6	Data science	423
7	SQL	399
8	Databases (relational, other)	375
9	Problem solving	372
10	Defining requirements	362
11	Presentation skills	227
12	Statistics	218
13	Computer science	211
14	Data modeling	195
15	Python (programming language)	189
16	Microsoft Excel	187
17	Business intelligence (BI)	179
18	Key performance indicators	179
19	Requirements gathering	165
20	Business insights	158
21	Knowledge acquisition	148
22	Forecasting	141
23	Business process	131
24	Extract transform load (ETL)	130
25	Project management	123
26	Data models	111
27	Data mining	108
28	Business process improvement	108
29	Data quality	107
30	Teamwork	101

The results show that the most demanded skills for business analytics jobs are related to data analysis, communication, visualization, data science, business requirements, and problem-solving. These skills reflect the core competencies of business analytics professionals, who need to be able to collect, process, analyze, and present data in various formats and contexts. These skills also align with the learning outcomes of many business analytics programs, which aim to equip students with the knowledge and skills to perform data-driven decision-making.

The analysis of the LinkedIn job postings reveals some interesting insights into the skills that employers are looking for in business analysts. In Table 1, we show the frequency of skills demanded for business analytics jobs. Please note that the numbers in this table are slightly different from other tables in this article, as these numbers report skills rather than

jobs. Sometimes a job will list a particular skill twice (e.g., “Analytics” and “Analysis”).

Of course, “business analytics” and “data analytics” are the most commonly listed skills for business analytics jobs on LinkedIn, with 1651 mentions. But this merely states the obvious: one must have analytics skills for an analytics job. Therefore, we turn our attention to the other skills in high demand for business analysts.

The most demanded skill for business analysts—aside from the obvious “analytics” skill—is strong communication skills. This is shown to be even more important if we include “presentations” (227 mentions) with “communication” (828 mentions). This indicates that business analysts are expected not only to analyze data, but also to communicate their findings and recommendations effectively to various stakeholders, such as managers, clients, and peers. Our results thereby reinforce earlier research by Wixom et al. (2014) and Radovilsky et al. (2018) that identified communications skills as foundational and highly sought after.

Visualization can be considered a different type of communication skill. A picture is worth a thousand words, and maybe ten thousand numbers. Visualization is a powerful communication tool that can help business analysts convey complex and nuanced information in a clear and concise manner. Visualization skills include the ability to create and interpret charts, graphs, dashboards, and other graphical representations of data. Visualization also requires the knowledge of various tools and software, such as Tableau, Power BI, and Excel, that can facilitate the creation and dissemination of visualizations. Visualization was mentioned 606 times in the job postings, making it the third most common skill after analytics and communication.

A facility with databases is a requirement of most analytics jobs. After all, one must be able to access the data before visualizing or analyzing it. Databases are organized collections of data that can be stored, retrieved, and manipulated using various methods and tools. One of the most common tools for working with databases is Structured Query Language (SQL), which is a programming language that allows users to query, manipulate, and analyze data stored in relational databases. SQL was mentioned 399 times in the job postings, making it the seventh most frequently occurring skill. Our “Databases” category includes “Data Warehousing,” and if we combine SQL with this broad “Databases” category, the 774 mentions would make it the third most demanded skill. The importance of databases is even more clear if we include “extract transform load (ETL)” in the list of database skills. ETL is a process that involves extracting data from various sources, transforming it into a suitable format, and loading it into a target database or data warehouse. ETL was mentioned 130 times in the job postings, indicating that business analysts are often involved in preparing and integrating data from multiple sources for analysis. In total, databases and database skills combine for 904 mentions. Clearly, any analytics curriculum should contain a very strong database component.

Another skill that is essential for business analysts is programming. The process of rigorously analyzing data usually



involves writing custom scripts or code, and programming skills are implicit in many of the skills listed in Table 1. For example, one simply cannot *do* “Analytics (Bus, Data),” the most frequently mentioned skill (1651 mentions), “Data Science” (423 mentions), or “Computer Science” (211 mentions), without simultaneously doing programming. Several programming languages are also mentioned explicitly. SQL is the most frequently mentioned programming language (399 mentions). Python is the second most mentioned programming language, mentioned 189 times. Computer science’s high rank, higher than Python, indicates that employers may be flexible as to the specific programming languages they require. Thinking algorithmically may be more important than knowing syntax.

Thus, to our eyes several key skills clusters emerge for business analysts: (1) *analytics*, including statistics, forecasting, machine learning, and data science; (2) *communication*, including visualization and presentations skills; (3) *programming*, including SQL, Python, and Excel; and (4) *databases*, including SQL. These skills clusters reflect the core competencies that business analysts need to succeed in the data-driven economy. They also provide a useful framework for designing and evaluating analytics curricula and courses. It is our contention that strong analytics programs will contain a mix of courses that, in the aggregate, equip students with this list of skills.

It is surprising that terms such as “artificial Intelligence,” “AI,” “generative AI,” “machine learning” and the like did not appear among the most desired skills for analytics jobs. This may illustrate the incredibly dynamic nature of the analytics field. At the time of writing (mid-2024), these terms are ubiquitous, but they were scarcely mentioned at the time of data collection toward the end of 2023. Alternatively, it may be that there is a significant gap between “data analysts” and “data scientists,” with the former emphasizing the analysis of business trends and metrics, and the latter emphasizing the creation of new business products and processes.

4.2 | Data visualization: Tableau versus Power BI

Data visualization is the process of transforming data into graphical representations that can communicate insights and trends effectively. Data visualization skills are essential for business analysts, who need to present and explain their findings to various stakeholders. However, there are many different tools and platforms for creating data visualizations, such as Tableau and Power BI. Which ones are more in demand and more valuable for aspiring analysts?

To answer this question, we restrict our sample to the 606 analytics job postings that require some form of data visualization skills; jobs that include the keywords “visualization,” “Power BI,” “Tableau,” “viz,” “dashboards,” and “dashboarding.” We then enumerate the jobs based on whether they explicitly mention Tableau or Power BI as a desired skill or not. The results are summarized in Table 2.

TABLE 2 Visualization jobs by Tableau and Power BI skills.

Power BI			
Tableau	No	Yes	Total
No	457	48	505
Yes	98	3	101
Total	555	51	606

Table 2 indicates that the majority of visualization jobs—457 out of the 606—require a general skill in “visualization” without further specification. Employers seem to value the ability to create effective and informative visualizations regardless of the specific tool or platform used. Tableau (101 mentions), however, is far more demanded than Power BI (51).

Dong and Triche (2020) found that Tableau is 10 times more important than Power BI (23% vs. 2%). Earlier research by Radovitsky, et al. (2018) and Dong and Triche (2020) found a preference by employers for Tableau over Power BI and over general visualization skills. Likewise, in their study of analytics market in India, Umamaheswaran et al. (2023) reported a specific demand for Tableau. Our analysis reveals this not to be the case in the United States. We find that employers in the United States currently prefer Tableau over Power BI. However, demand is for visualization skills in general, rather than in a particular platform. As visualization is seen by industry as a general skill, switching between platforms can be relatively easy. All job applicants, however, must understand the principles of effective data visualization. Students looking to self-study or instructors who are trying to construct a curriculum are advised to focus on learning the general principles and techniques of data visualization and to use Tableau as a convenient and powerful tool for creating and presenting visualizations.

4.3 | Programming languages

In this section, we analyze the data collected from the job postings to answer the research question: which programming languages should analytics majors learn? We use cross-tabulations to compare the frequency and co-occurrence of different programming languages in the job requirements. The results are summarized in Table 3 below, which shows the count of job postings that mention each programming language.

Which programming languages should analytics majors choose? Table 3 shows that SQL is the most vital programming skill for a budding analyst. (The astute reader will notice that the frequency of SQL in Tables 1 and 3 differs slightly. This is because the unit of analysis in Table 1 is skills, whereas in Table 3 it is jobs. There were a small handful of job postings where SQL was mentioned more than once, in slightly different wordings, in their list of desired skills. For example, some job postings list “SQL” along with



TABLE 3 Jobs requesting specific programming languages.

Language	Count
SQL	385
Excel (or VBA)	288
Python	189
R	98
SAS	83
Java	28
Statistical programming	14
Stata	3
SPSS	1

“SQL Server.”) SQL is essential for accessing and manipulating data from various sources, such as relational databases, data warehouses, and cloud platforms. Thus, our results reinforce earlier work by Wixom et al. (2014), which found that SQL is the most important foundational technical skill for business analysts.

After SQL, Excel and VBA is the second most demanded software skill. Excel is a widely used tool for data analysis, reporting, and visualization, and it offers many built-in functions and features that can simplify and automate common tasks. However, the level of proficiency required in Excel may vary depending on the job. For example, some jobs may require programming Excel using VBA, which is a higher-order skill than merely using Excel’s basic functionality, such as pivot and lookup tables.

A perennial question is whether students should specialize in Python or R, which are both popular and powerful programming languages for data analysis, machine learning, and statistical modeling. Dong and Triche (2020) found that Python and R had the same relative importance (each showing up in roughly 11% of job postings). We find, however, that Python now dominates R; Python jobs outnumber R jobs two to one. Python is more versatile and flexible than R, as it can be used for a variety of purposes, such as web development, automation, and scripting. However, this does not mean that R is irrelevant or inferior to Python. R is widely used in the academic and research communities, and it offers many specialized packages and functions for advanced statistics and data visualization. Therefore, students may benefit from learning both Python and R, depending on their career goals and interests.

Some jobs might prefer a proficiency in more than one programming language. In Table 4, we restrict our sample to jobs that list Python, R, or SQL skills, and examine whether there is overlap in the demand for these three languages. Table 4 verifies that SQL, a database language, is not a substitute for programming languages, such as Python and R. In fact, SQL and Python seem to be complementary skills, as 105 of the 385 jobs that mention SQL also mention Python. This suggests that employers are looking for candidates who can combine data extraction and manipulation skills with data analysis and modeling skills. On the other hand, SQL and

TABLE 4 SQL, Python, and R.

SQL			
R	No	Yes	Total
No	64	368	432
Yes	81	17	98
Total	145	385	530
SQL			
Python	No	Yes	Total
No	61	280	341
Yes	84	105	189
Total	145	385	530
Python			
R	No	Yes	Total
No	271	161	432
Yes	70	28	98
Total	341	189	530

TABLE 5 Jobs requesting Python or its packages.

Variable	Count
Python	189
Matplotlib	16
Pandas	12
TensorFlow	10
NumPy	4
Seaborn	4
PyTorch	3
SciPy	2
Keras	1

R seem to be less compatible, as only 17 jobs mention both languages. This may reflect the fact that SQL is more suited for business, while R is more suited for academic research.

In conclusion, our analysis of the data suggests that analytics students should study SQL, Excel, and Python as the core programming skills for their careers. They may also consider learning R as an additional skill if they are considering an academic or research-oriented career. More traditional statistical programming languages, such as SAS, Stata, and SPSS, are of limited use for the aspiring analyst.

4.4 | Python packages

Given that Python is one of the more sought after programming languages for Business Analysts, are there specific packages that are more valuable? According to the data in Table 5, the answer seems to be “no.” Of the 206 jobs that list Python or one of its packages, fewer than 10% specifically ask for Matplotlib, Pandas, or TensorFlow. Either employers do not explicitly require a further specialization in a Python



TABLE 6 The number of jobs containing skills in various fields.

Field	Freq.	Percent
Data science	423	16%
Statistics	306	11%
Finance	361	13%
Computer science	211	8%
Marketing	147	5%
Economics	82	3%
Accounting	81	3%
All jobs	2676	

Note: The entries represent partially overlapping subsets of 2676 jobs; therefore, the reported percentages are not expected to sum to 100%.

package or job applicants who need to know Python are implicitly expected to be familiar with the more common Python packages.

4.5 | Compatibility with other fields

Research by Deng et al. (2016) implied that hiring firms want well-rounded business analysts: business-school trained analysts with skills in algorithms and programming, and computer science and statistics graduates who also understand business strategy. Likewise, Wixom et al. (2014) reported that business knowledge is the fifth most valued foundational job skill.

This leads us to our next question: which business fields (other than business analytics itself) are most compatible with business analytics jobs? Or, to put it another way, in which other departments should analytics students take classes?

To answer this question, we classify each job based on the various keywords in its skills list. Here, we consider seven such fields: accounting, economics, finance, marketing, data science, statistics, and computer science. For example, if a job has a skills list that contains the word “accounting” or “accountant,” it is considered a job associated with the accounting field. Jobs with skills lists that contain some variant of the word “finance” or “financial” are classified as finance jobs, and so forth. (For finance, we also included “investing” and “banking.”) Some skills may belong to more than one category, such as “financial accounting”; likewise, jobs may be associated with more than one field. Table 6 reports the number of jobs containing the various keywords. Since fields are not mutually exclusive, the reported percentages do not need to sum to 100%.

Table 6 shows that the most common fields for business analytics jobs are not always part of the traditional business curriculum. Data science is the most commonly listed supplementary field for analyst jobs, comprising 16% of the jobs. This indicates that business analytics jobs require a strong foundation in regression, classification, and analyzing unstructured data. Statistics is the second most occurring major, showing up 11% of the time. Computer science ranked as the fourth most important major complementary to business analytics jobs. These majors provide the theoretical and practical knowledge of statistical methods, models, and

TABLE 7 Most frequently occurring finance skills.

Finance skills	Freq.
Financial analysis	76
Finance	73
Financial performance	33
Banking	27
Oracle Financials	26

TABLE 8 Most frequently occurring marketing skills.

Marketing skills	Freq.
Marketing	47
Marketing analytics	39
Digital marketing	27
Marketing strategy	16
Marketing campaigns	11

algorithms that are essential for business analytics, as well as the programming and coding skills that are needed to implement them.

Among business majors, Table 6 shows that finance is the field most complementary to business analytics jobs; 13% of all job postings contain at least one financial skill. It more than doubles the popularity of the next-most favored business field, marketing, which is mentioned in only 5% of the jobs.

Given the results above, if a business analytics student is considering taking additional business classes, they should consider finance or marketing. If they are bold enough to venture outside of their college, data science or statistics classes would be very beneficial.

To further explore the compatibility of the various majors with business analytics jobs, we dig deeper, examining the specific skills associated with finance and marketing, the two most compatible business-oriented majors for analysts.

Table 7 lists the top five financial skills listed among business analytics jobs. Financial skills are defined here as any skill including the words “finance” or “financial” (excluding “financial reporting,” which is an accounting skill), as well as “banking,” “invest,” “investment,” and “investing.” The frequencies in Table 7 show the number of times a particular financial skill was listed; thus, the unit of analysis is the skill, not the number of jobs listing the skill. The most commonly desired skills are also the broadest or most general (e.g., “finance” or “financial analysis”). Oracle Financials is a software system for automating a company’s financial management functions.

Table 8 lists the top marketing-related skills for business analytics jobs. Of these, general marketing ranks as the most important, followed by marketing analytics. This much is to be expected for business analytics jobs. Digital marketing is the third most commonly listed marketing skill. Digital marketing jumps to first place (with 66 mentions) when we include synonyms and subcategories such as email marketing (9), marketing automation (5), social media marketing (4), digital marketing channels (3), online marketing (3), cloud marketing (2), Salesforce Marketing Cloud (2), search



engine marketing (2), TikTok marketing (2), and influencer marketing (1).

4.6 | Cluster analysis

For the final portion of our analysis, we conducted a cluster analyses of the skills in the LinkedIn job postings in our dataset.

The first cluster identified centers around “Business Analysis and Process Improvement.” It emphasizes the importance of skills associated with business process improvement, enterprise resource planning, business systems analysis, and facets of the software development life cycle.

In the second cluster, “Advanced Technical and Data Science Skills,” we encounter advanced technical skills crucial for modern business analytics: proficiency in Python, ETL, data science, cloud computing platforms (Amazon Web Services, Microsoft Azure), machine learning, SQL, and big data. These skills indicate the growing demand for professionals adept at handling intricate data processing and analyses.

The third cluster, “Financial and Strategic Analysis,” illustrates the importance of financial analysis, budgeting, accounting, strategic initiatives, and market analysis. This cluster suggests that a robust understanding of financial aspects is fundamental for effective decision-making.

The fourth cluster, “Tools and Office Software,” encapsulates skills related to commonly used tools and office software, including Microsoft Office suite of applications. These tools are ubiquitous in today’s business world.

“Data Modeling and Leadership,” the fifth cluster, groups skills such as data modeling, data architecture, team leadership, and data and project facilitation. This cluster highlights the leadership qualities and organizational skills required for managing data-related projects in the business analytics domain.

The sixth cluster, “Technical Expertise,” is characterized by specific technical skills such as VBA, Oracle Database, SAP, and Microsoft Power Query. These skills underscore the importance of technical expertise in specialized software and database technologies.

The seventh cluster highlights the significance of “Structured Planning and Reporting” in the realm of business analytics. It groups planning-related skills, including business and strategic planning, and graphic design principles.

The final cluster, “Economics and Banking,” centers around skills related to economics, banking, Oracle Financials, and business profitability. This implies a connection between business analytics and the financial sector, emphasizing the relevance of financial knowledge in analytics roles.

In summary, the analysis of clusters offers insights into the diverse and interconnected skill sets integral to success in the field of business analytics. These findings, in turn, have implications for curriculum design, as we discuss in the next section.

5 | RECOMMENDED CURRICULUM

The findings of this study have several implications for designing the business analytics curriculum. The curriculum should provide a balanced and comprehensive coverage of both technical and nontechnical skills. It should teach how analytics can be used to benefit various business domains and functions. The curriculum should be flexible and adaptable to the changing demands of the job market for business analysts. It should also be integrated with experiential learning opportunities, such as projects, internships, and competitions, to enhance a student’s practical skills and employability. Based on the analysis above, we propose the following comprehensive curriculum as a way to equip students with the necessary knowledge and skills for a successful entry into the field of business analytics:

First, some topics are fundamental and should be taught to all business students, not just those majoring in business analytics. Given the importance of cross-functional domain knowledge, all analytics students should be required to take the full component of core business classes. These should include at least one class in accounting, economics, finance, marketing, and management. In other words, we recommend that analytics be taught in a business school, rather than an engineering or A&S school.

5.1 | Tools and office software

Students require proficiency in commonly used productivity software such as the Microsoft Office suite of applications (Excel, PowerPoint, Outlook). Mastery of these tools is essential for presentational business employees, and these tools should be taught throughout the business curriculum.

5.2 | Introduction to business statistics

Every student in a business college should take an introductory course in business statistics. This course should expose student to the fundamentals of descriptive, prescriptive, and predictive analytics.

Next, we turn to the courses that are specific to business analytics students.

5.3 | Visualization

We recommend at least one course dedicated to visualization. This course should expose students to the basic principles of proper graphing, the psychology and physiology of visual perception, and to at least one modern platform (either Tableau or Power BI) used for creating dynamic visualization dashboards. Camm et al. (2023) provide some suggested features for visualization classes, particularly at the graduate level. They propose several different alternatives, depending on whether the goal is the consumption or creation of visualizations. They recommend that either Tableau or Power BI be included in the creation-focused visualization classes.



5.4 | Communication

Communication skills are essential to success in business analytics. We recommend dedicating one whole course to the communication of statistical results. This class should train students how to write reports and give oral presentations that defend the design of their analyses and explain their statistical results to nontechnical audiences. Moreover, we recommend that every analytics course contain some component dedicated to statistical communication.

5.5 | Programming

Students should take two introductory programming classes using Python. In our experience, the first course is adequate for teaching the fundamentals of functional programming; the second should focus on object-oriented programming.

5.6 | Databases

We recommend one class on extracting, transforming, and loading data (ETL) onto and from modern databases. The course should teach the fundamentals of SQL programming. Such skills are crucial for students to access, manipulate, and integrate data effectively.

5.7 | Advanced statistics

We recommend three to four courses dedicated to topics in advanced statistics. These could include courses in cross-sectional econometrics, time-series econometrics or forecasting, causal analysis, geospatial data (GIS), simulation, machine learning, or data science.

5.8 | Capstone project

Finally, we recommend a capstone class dedicated to performing one larger-scale data-based project. This experiential learning project may involve a partnership with a local company, an internship, or possibly an analytics competition, all of which can enhance a student's practical skills. More importantly, it will allow students to develop a showpiece to share with potential employers, demonstrating their hard and soft skills. We recommend that a large component of the course grade involve the written and oral communication of the statistical results and business recommendations flowing from them.

This comprehensive curriculum addresses the diverse skill sets identified in the analysis, providing students with a well-rounded education in business analytics. It develops students' technical proficiency, communication skills, and domain-specific knowledge. Additionally, it aligns with the learning outcomes sought by employers and reflects the current demands of the business analytics industry.

6 | CONCLUSIONS

This article identified the most in-demand skills for business analytics jobs based on a content analysis of 2676 online job postings. The article also discussed the implications of these findings for curriculum design and career development, knowledge that provides valuable insights into the complex landscape of this dynamic field. Identifying and categorizing skills was necessary to navigate the nuances present in job postings, where synonyms and variations abound. We employed a meticulous approach, combining similar skills and addressing the subjectivity inherent in this process.

Table 1 offers an overview of the most demanded skills for business analytics jobs. The results underscore the prominence of data analysis, communication, visualization, data science, and problem-solving skills. These core competencies align with the expectations of business analytics professionals, who must adeptly collect, process, analyze, and present data across diverse contexts.

Effective communication—both written and oral—is a vital job skill. Thus, our research supports the finding by Wixom et al. (2014) that industry places the highest value on communication skills. A student's ability to effectively communicate statistical information should be developed in a single dedicated class on statistical communication. However, given its importance, every course in the analytics curriculum should have some communication component.

Visualization skills emerged as a powerful communication tool. While visualization is only required in 24% of analytics undergraduate programs (Ceccucci et al., 2020), our results indicate that visualization is a defining feature of modern business analytics jobs. Thus, there seems to be a gap between what schools and businesses require. We argue that every analytics curriculum should contain a strong visualization component.

Databases, particularly SQL, took center stage as a foundational requirement for accessing and manipulating data. This, too, supports earlier results by Wixom et al. (2014), where SQL was the second most valued foundational skill. Two-thirds of undergraduate analytics programs require a databases course (Ceccucci et al., 2020). We argue that every analytics program should teach databases and SQL.

In a specialized analysis of data visualization skills, we explored the demand for Tableau and Power BI. The results indicated that while Tableau is a preferred tool, the overarching demand is for a general visualization skill. This nuanced understanding guides our recommendation for students and instructors to focus on learning the general principles of data visualization, using Tableau as a valuable tool in this context.

Programming, especially Python and SQL programming, also emerged as a vital skill cluster.

Dong and Triche (2020) documented an equally high—and growing—demand for Python and R. We find, however, that Python now overwhelmingly dominates R.

The analysis also uncovered a complementarity between Python and SQL, indicating a preference among employers for candidates proficient in both. Traditional statistical



programming languages such as SAS, Stata, and SPSS exhibited minimal demand. Students and job seekers are encouraged to master Python and SQL, and schools to incorporate both into their curricula.

The research by Deng et al. (2016) implied that hiring firms want well-rounded business analysts: business-school trained analysts with skills in algorithms and programming; and computer science graduates who also understand business strategy. We identified finance and marketing as the most sought-after double majors for business analytics roles. Data science, statistics, and computer science also featured prominently, highlighting the interdisciplinary nature of business analytics. Accounting and economics, though less common, provided valuable domain knowledge.

Cluster analysis delved into specific skill clusters, revealing distinct emphases on business analysis, technical proficiency, financial and strategic analysis, tools and office software, data modeling and leadership, technical expertise, planning and reporting, and economics and banking. These clusters offer a nuanced understanding of the diverse skill sets integral to success in business analytics.

Drawing from these insights, our recommended curriculum aligns with industry demands and learning outcomes sought by employers. It covers key clusters, including analytics, communication, programming, databases, business analysis and process improvement, advanced technical and data science skills, financial and strategic analysis, tools and office software, data modeling and leadership, technical expertise, planning and reporting, and economics and banking.

In summary, our research equips students and educators with a comprehensive description of the skills necessary for success in business analytics. These recommendations offer a blueprint for curricular design and help align the skills of students and other job seekers with the current demands of the business analytics industry.

The limitations of this research are that it relies on the quality and accuracy of the job postings, which may vary depending on the source, the employer, and the writer. It also involves some subjectivity in combining similar skills into one category. Future research should compare the skills for business analytics jobs across different regions, industries, and sectors, to identify the variations and trends in the demand for business analytics skills.

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